

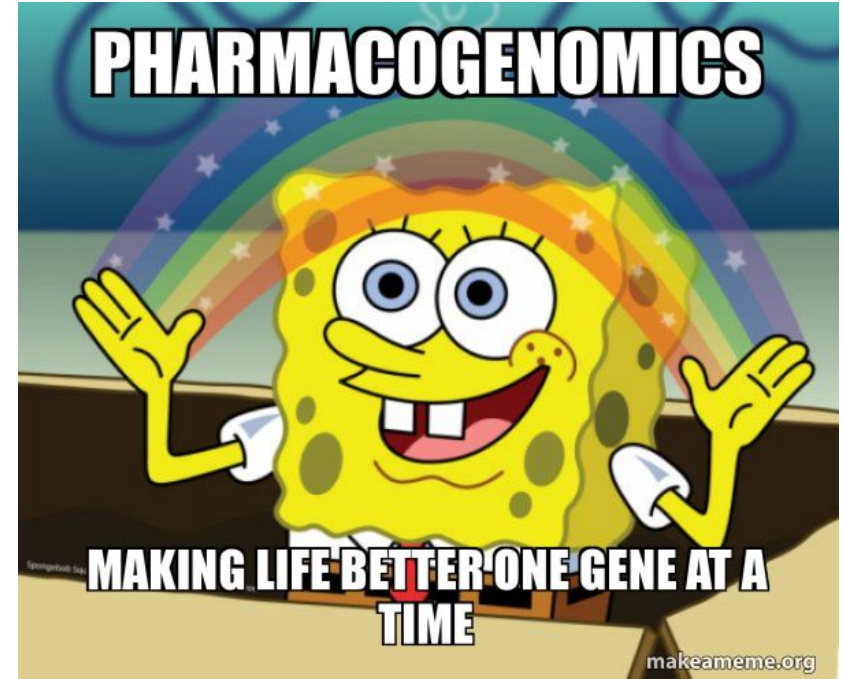
Applications III: Pharmacogenomics

Instructor: Caryn Geady, PhD

Date: 16th March 2026

Today's Objectives

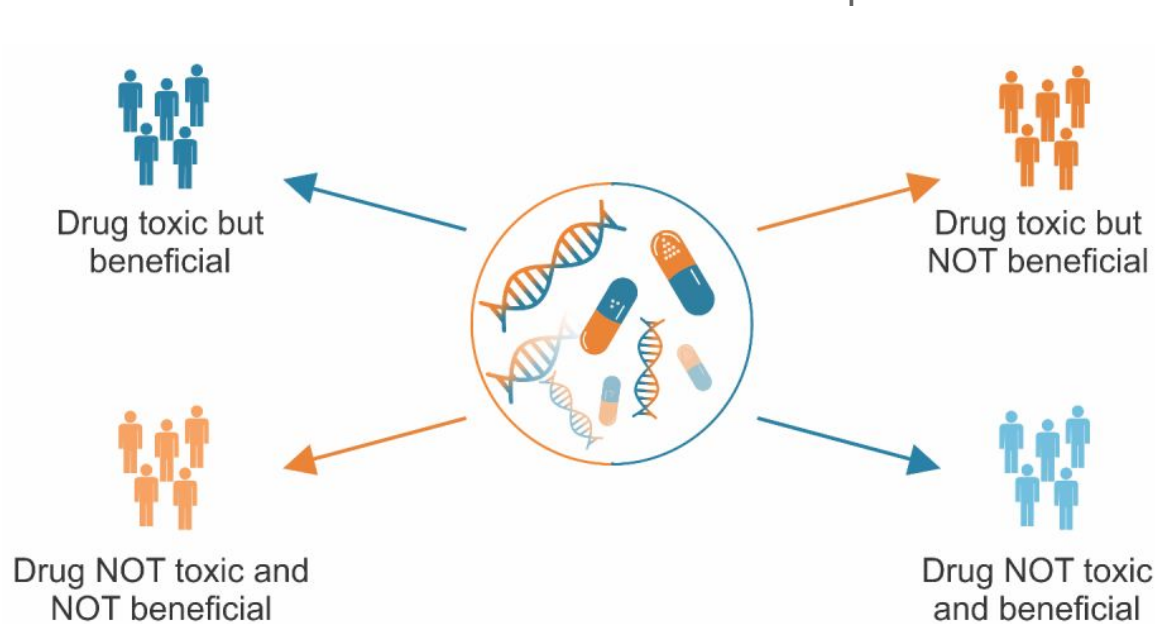
- Understand **Pharmacogenomics** and its impact on **personalized medicine**;
- Explore AI applications for **predicting drug response**;
- Learn about **key techniques** in pharmacogenomics.



Introduction to Pharmacogenomics

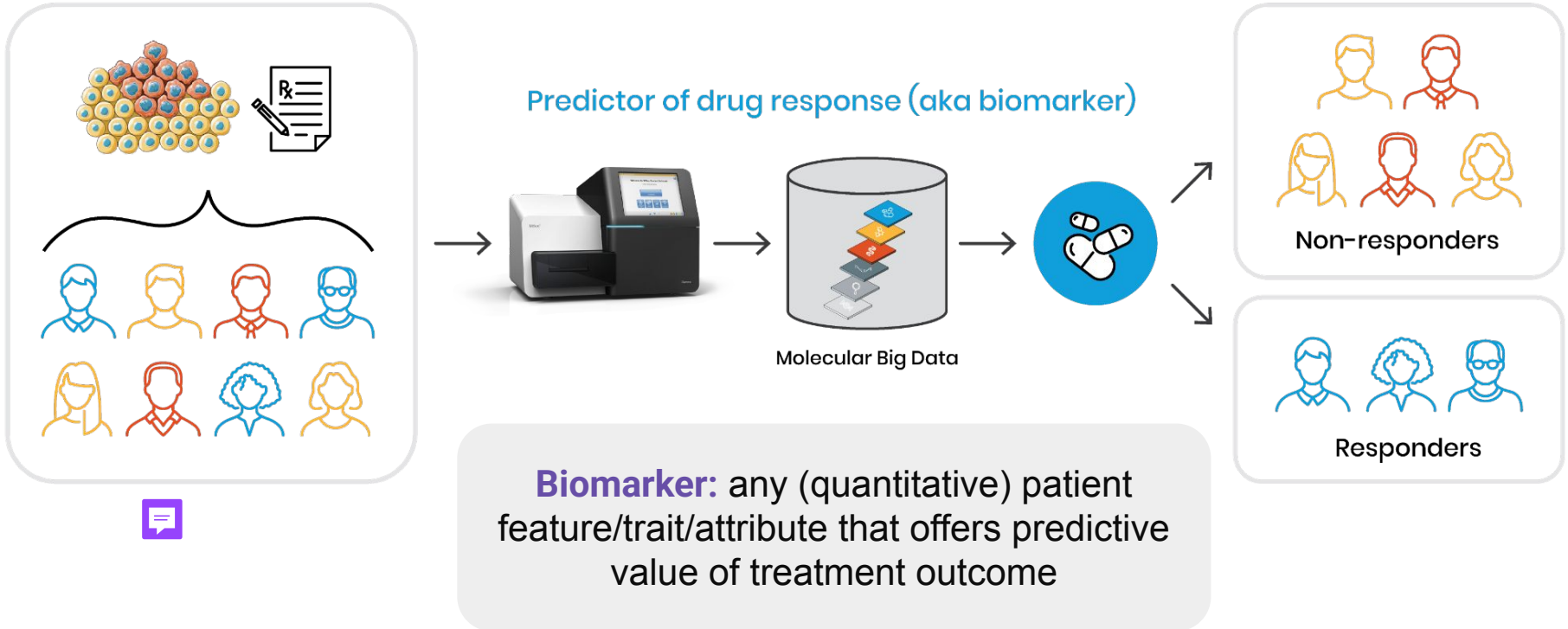
What is Pharmacogenomics?

- Study of how genetic variation affects an individual's response to drugs;
- **Role in precision medicine** → Tailored treatment plans.

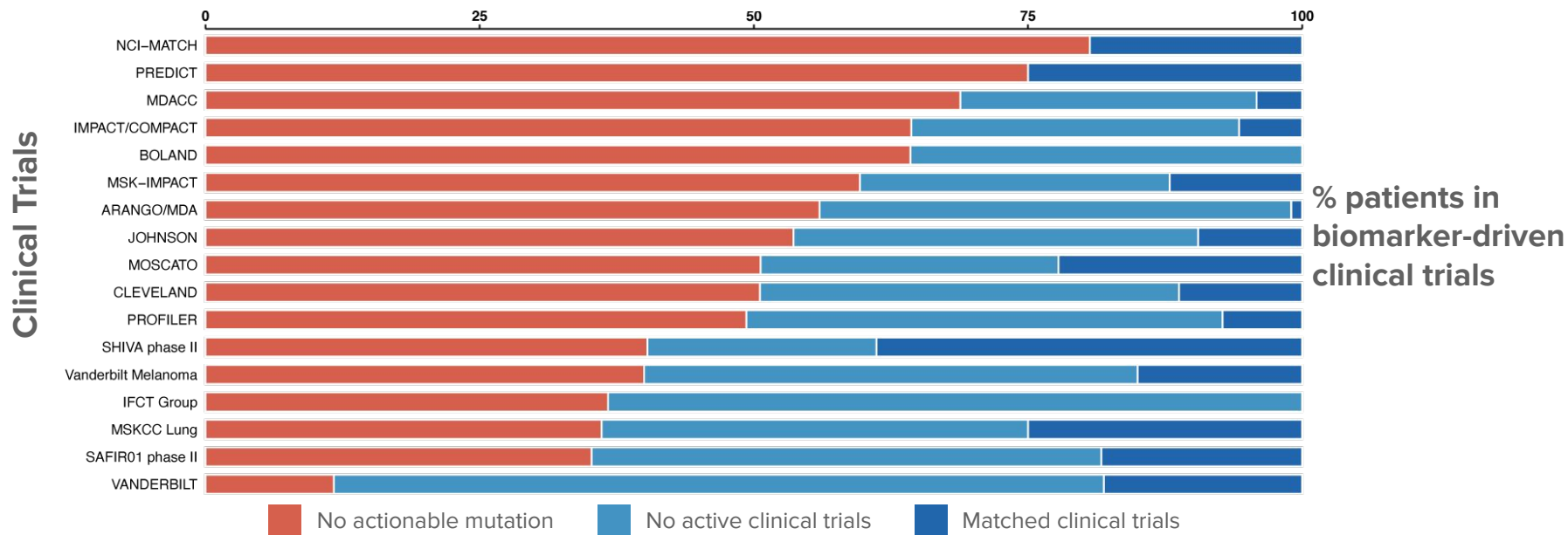


Precision Oncology and Biomarkers

All patients with same diagnosis



Basket/Umbrella (biomarker-driven) clinical trials



Despite large investments in major cancer centres, only ~50% patients could be matched to clinical trials based on targeted sequencing

Challenges for matching patients to therapies

- No access to drugs, not enough active clinical trials
- Small sequencing panels
- Profiling usually limited to a single assay (DNA, mRNA or protein)
 - **Very few known gene-drug associations (biomarkers)**

We barely scratched the surface!

Onc@KB

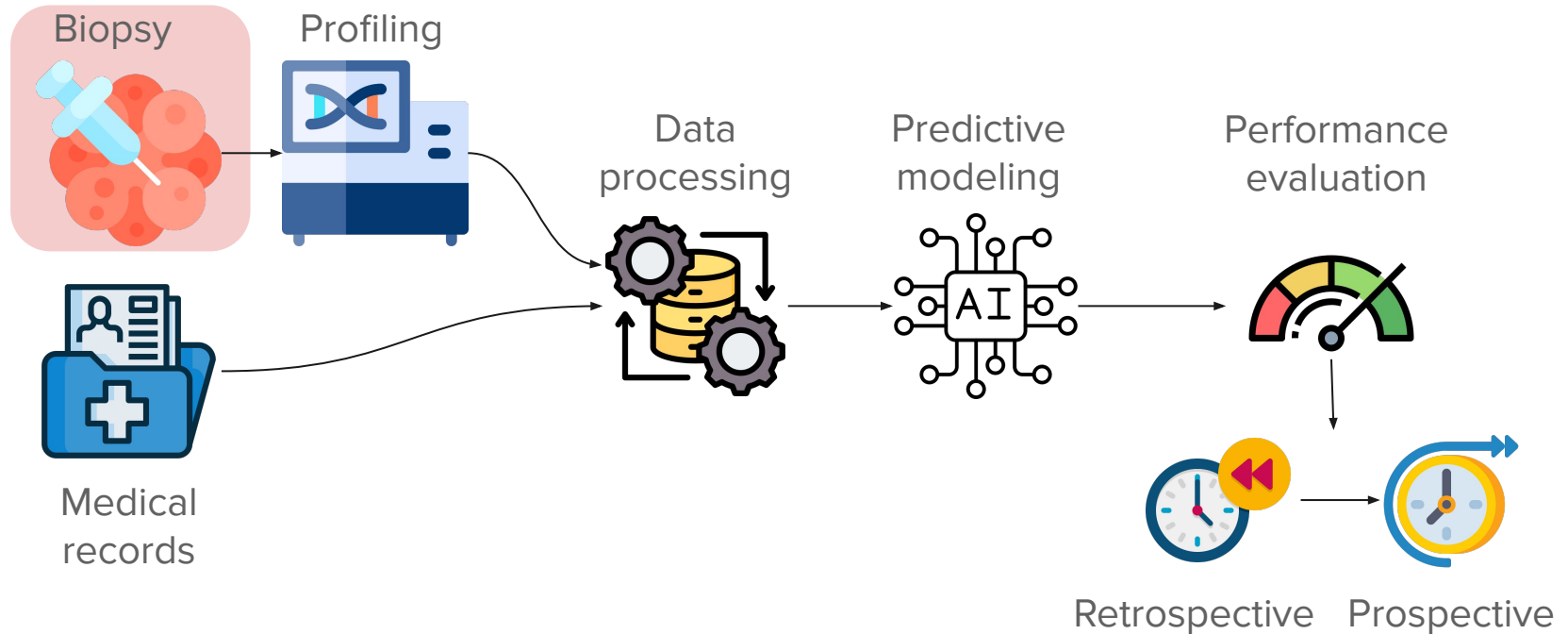
Only **79** genes out of ~26,000 genes have been associated with drug response
Only **135** drugs out of ~1000 anticancer drugs have an associated biomarker



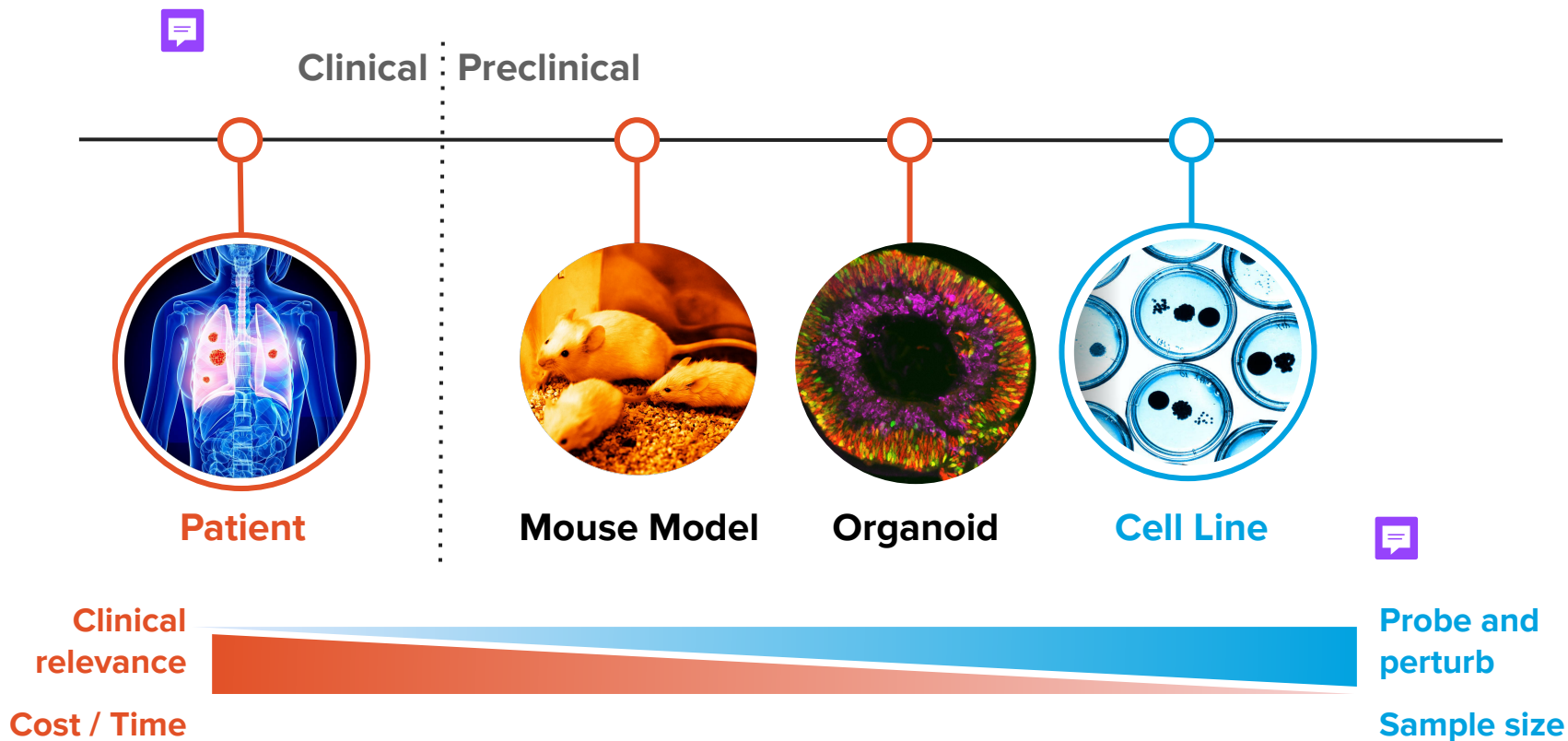
Need for new predictors of drug response

ML/AI for Drug Response Prediction

The Workflow for AI-Based Biomarker Discovery

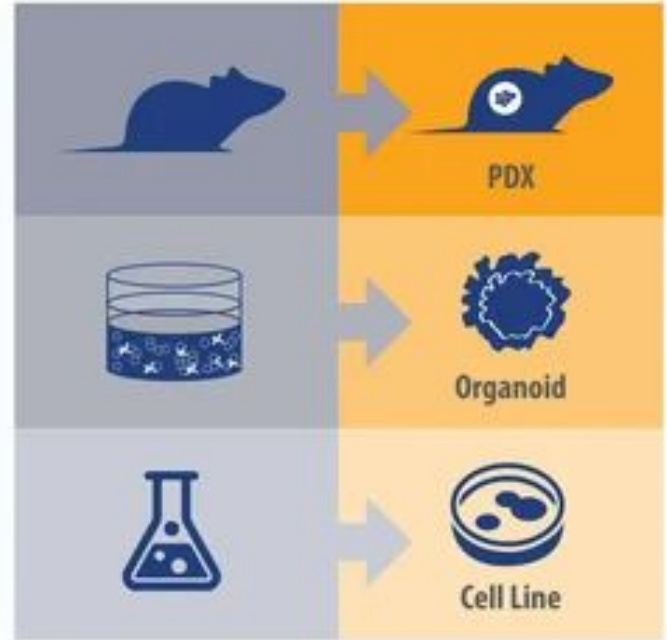


Pharmacogenomic Model Systems

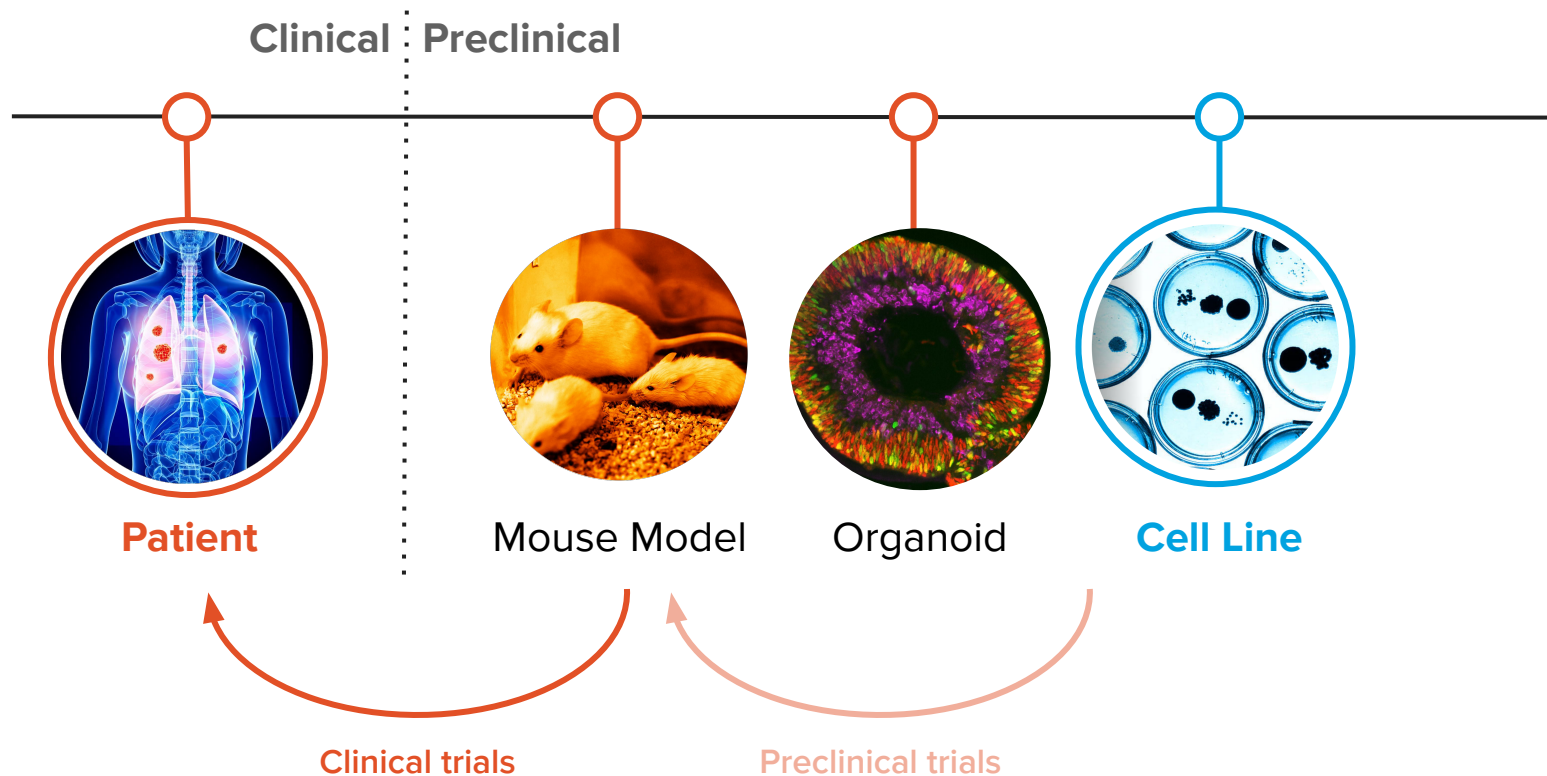


Preclinical Model Systems

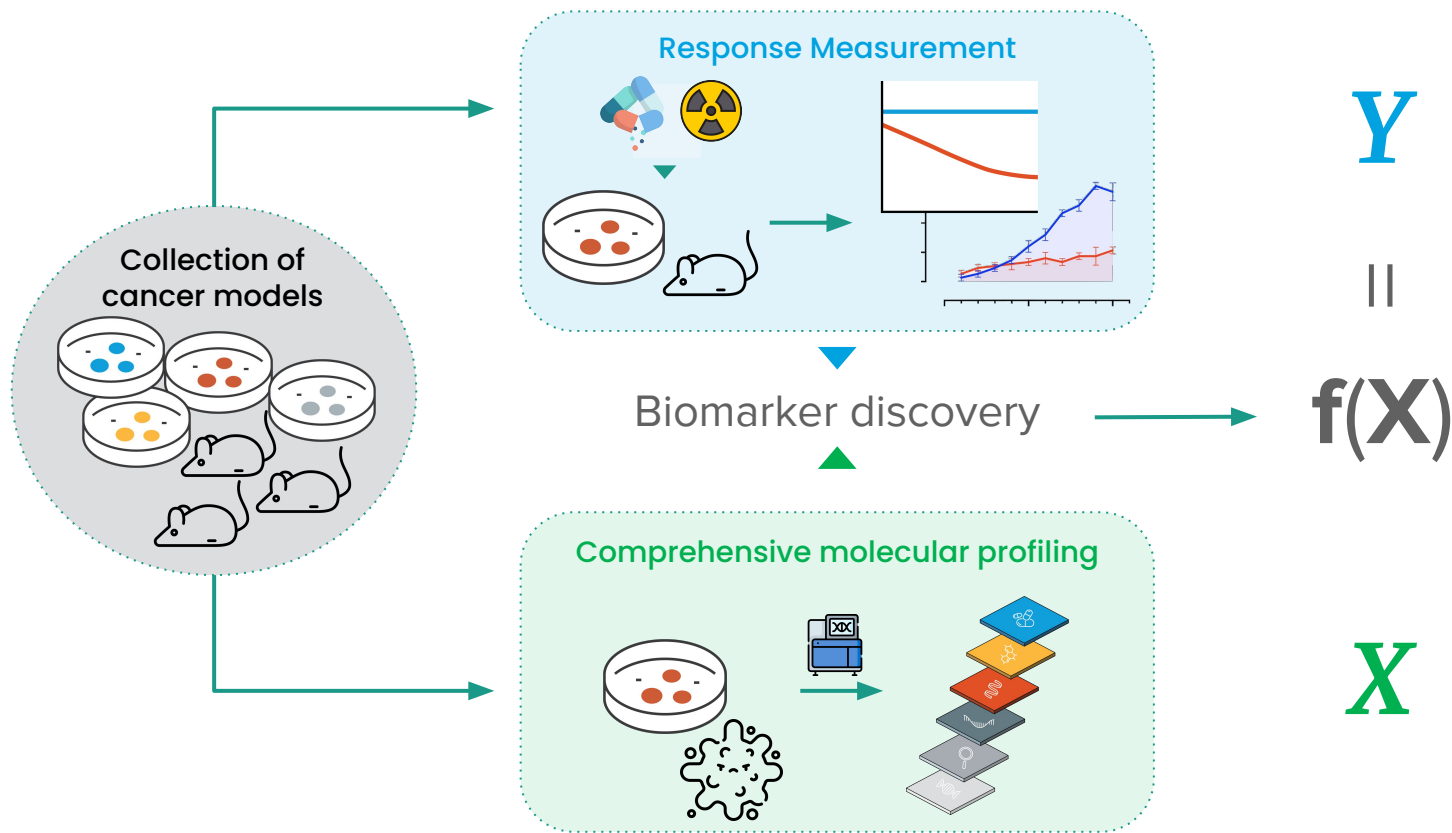
- Preclinical models **enable high-throughput profiling** +/- perturbations:
 - **Chemical perturbations:**
 - drug screening
 - **Genetic perturbations:**
 - sh/siRNA
 - CRISPR;
- ...preclinical models **do not necessarily recapitulate** the relevant features of patient tumors (and their many subtypes)



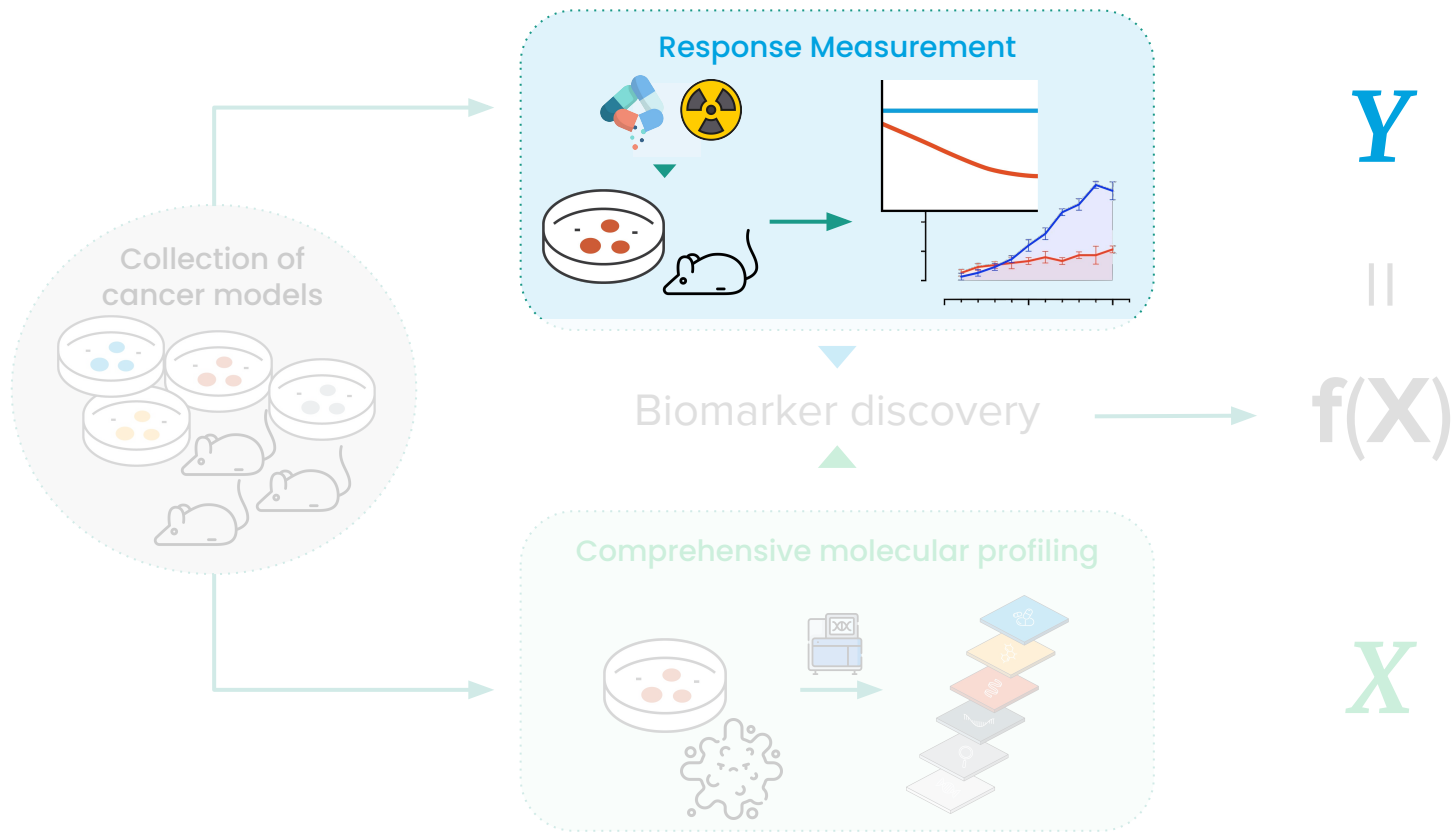
Clinical Validation



Data Components for Biomarker Discovery



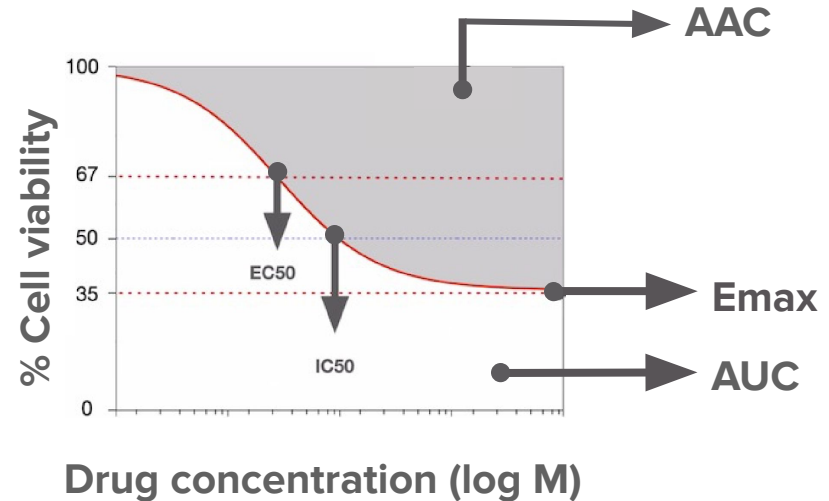
Data Components for Biomarker Discovery



Drug Response / Outcome Data

Metrics for quantifying effect of drug on cell viability:

- **IC50:** drug concentration to inhibit 50% of cell viability
- **EC50:** drug concentration to achieve 50% of cell viability
- **AUC/AAC:** area under/above dose response curve to represent drug effect on cell viability across all tested doses



Drug Response / Outcome Data

Metrics for quantifying effect of drug on tumor growth:



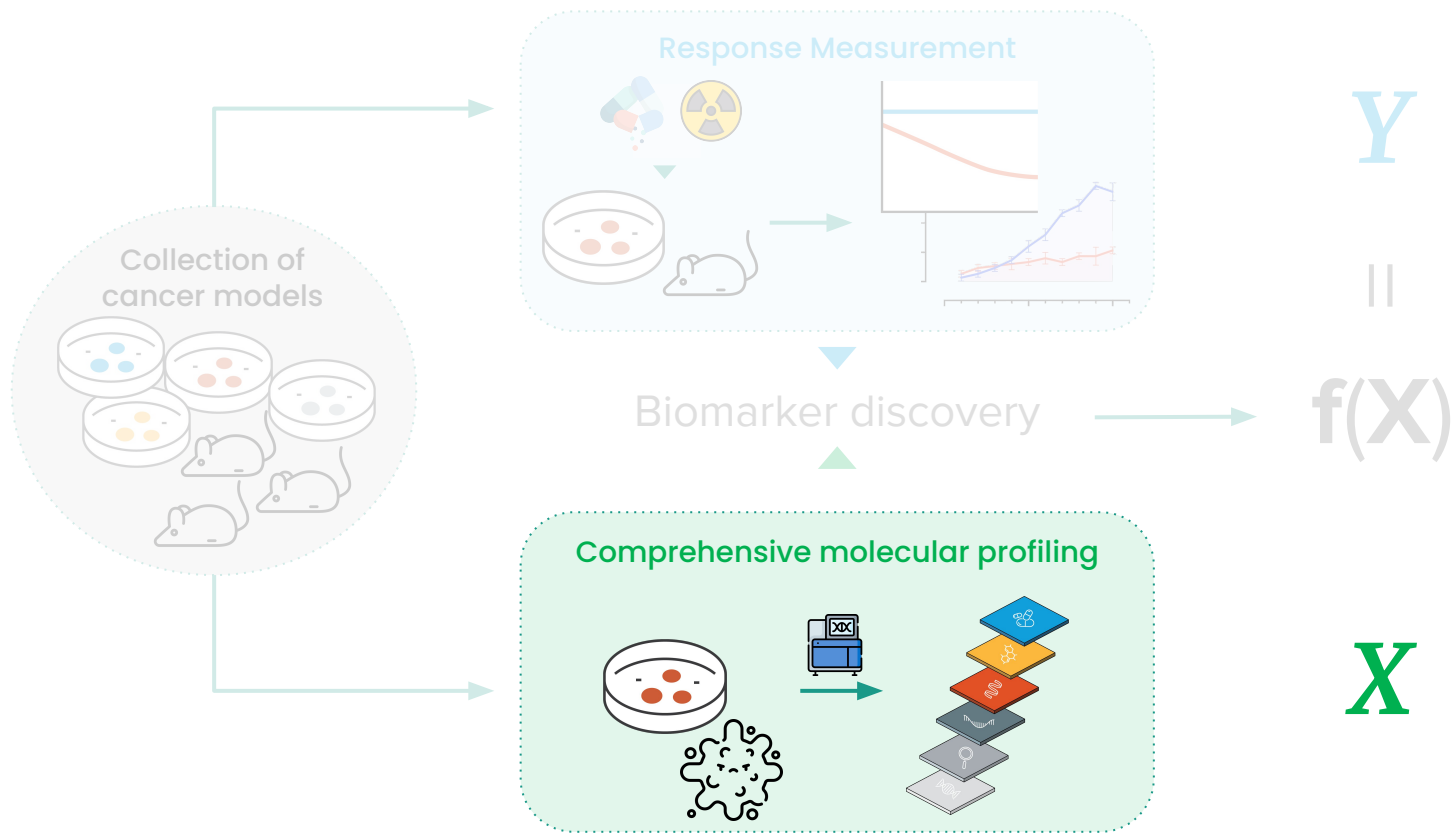
- **mRECIST:** outcome categories (definitions to the right)
- **AUC/AAC:** area under/above dose response curve to represent drug effect on tumor growth across all tested doses

mRECIST	Definition
CR	Disappearance of any intratumoral arterial enhancement in the target lesion
PR	>30 % decrease in the diameter of the viable target lesion ^a
SD	Any cases that do not qualify for CR, PR, and PD
PD	Increase of at least 20 % in the diameter of the viable target lesion ^a

mRECIST modified Response Evaluation Criteria in Solid Tumors, *CR* complete response, *PR* partial response, *SD* stable disease

^a Taking as reference the baseline diameter of the target lesion

Data Components for Biomarker Discovery



Data Modalities

- Multiple data modalities exist to train drug response predictors:
 - **Clinical** parameters (age, sex, tumor stage,...)
 - **Omics**: genomics, transcriptomics, epigenomics, proteomics,...
 - **Imaging**: radiology, pathology, microscopy,...
- Modalities that are not available in clinical settings can still be used to **inform** model development:
 - **E.g., ENLIGHT-DeepPT: Using histopathology to predict gene expression patterns for outcome prediction**

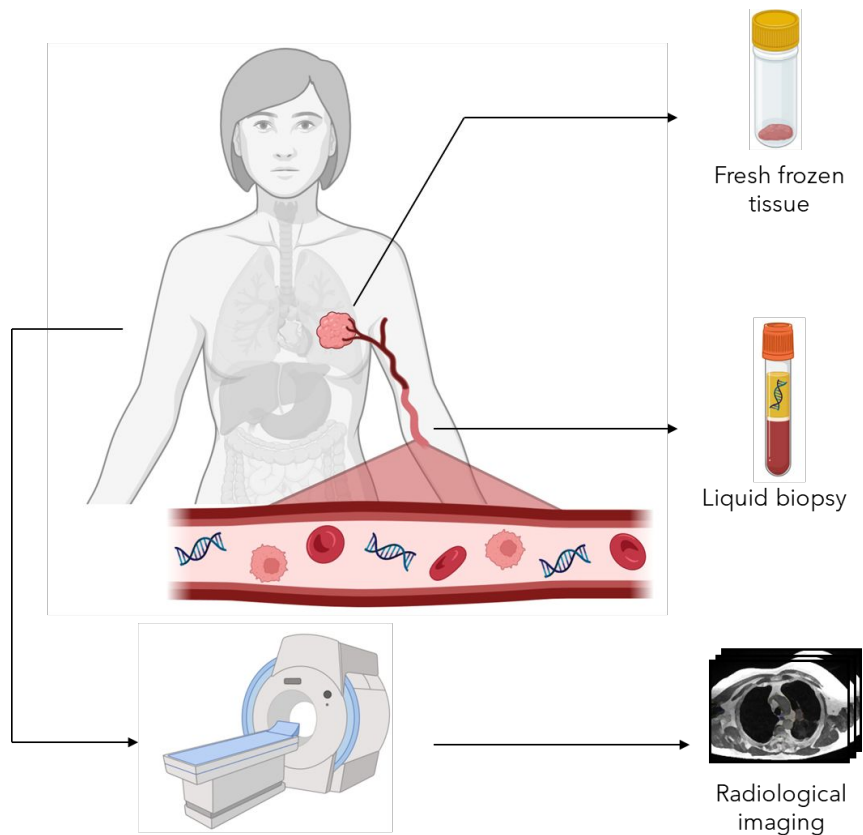
Prediction of cancer treatment response from histopathology images through imputed transcriptomics

 Sign in

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Methods for Patient Profiling



Tissue biopsy:

- For solid tumors
- Can be subjected to most sequencing procedures
- Can be inaccessible, invasive, more costly

Liquid biopsy:

- For solid and liquid tumors
- Limited molecular feature detection
- Less invasive and costly, serial collection, rapid turnaround

Radiological imaging:

- For solid and liquid tumors
- Radiomic features
- Least invasive

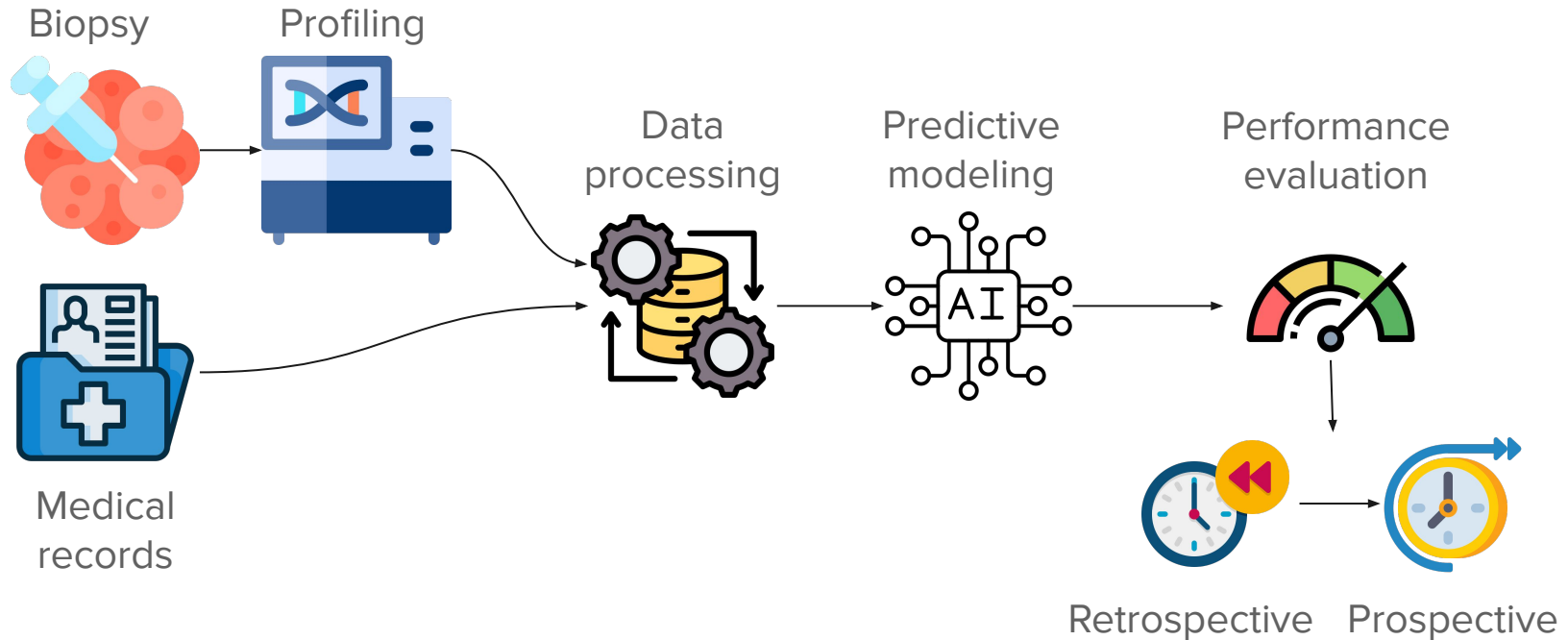
Most invasive



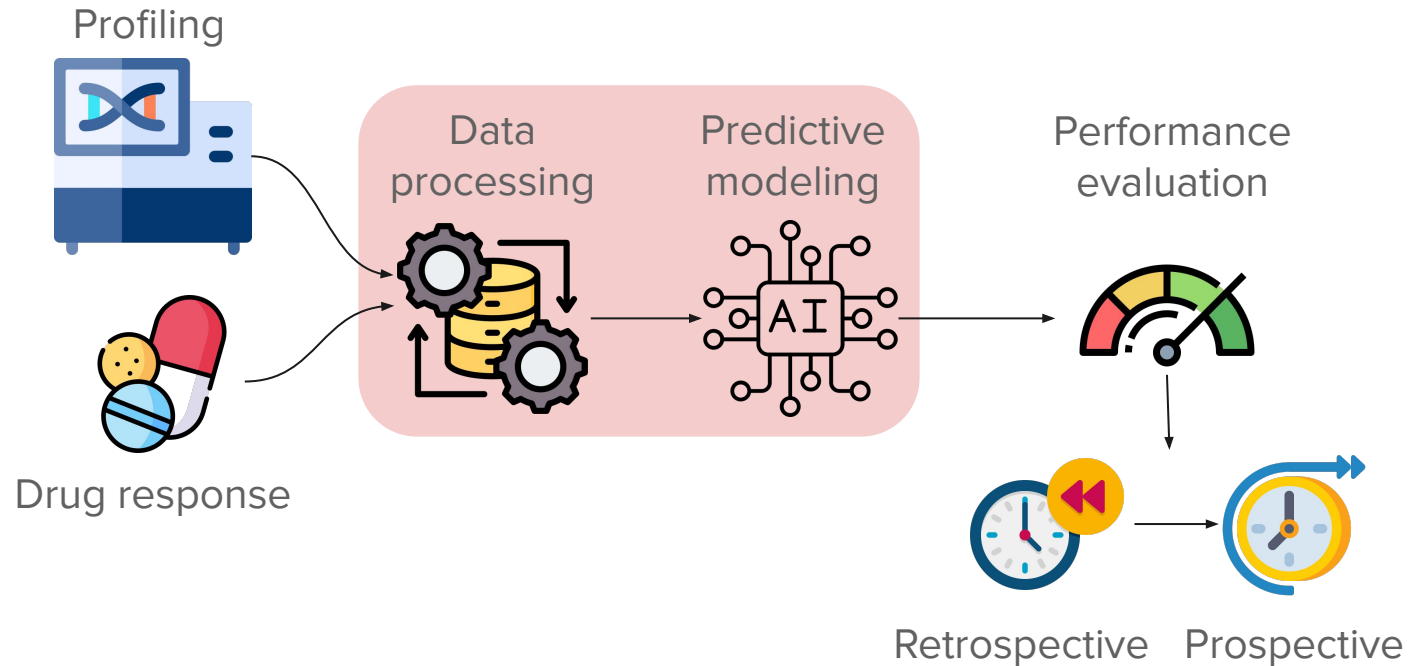
Least invasive

Key Techniques

The Workflow for AI-Based Biomarker Discovery



The Workflow for AI-Based Biomarker Discovery



Pharmacogenomics Analysis

- **Data Preprocessing:**
 - Quality control,
 - Normalization, and
 - Handling missing values.
- **Exploratory Data Analysis:**
 - Visualizing data distributions and relationships.
- **Statistical Testing:**
 - Identifying significant associations between genes and drug response.
- **Validation:**
 - Confirming findings in independent datasets.



Pharmacogenomics Data Analysis Walk-Through

Step 1. Data Preprocessing:

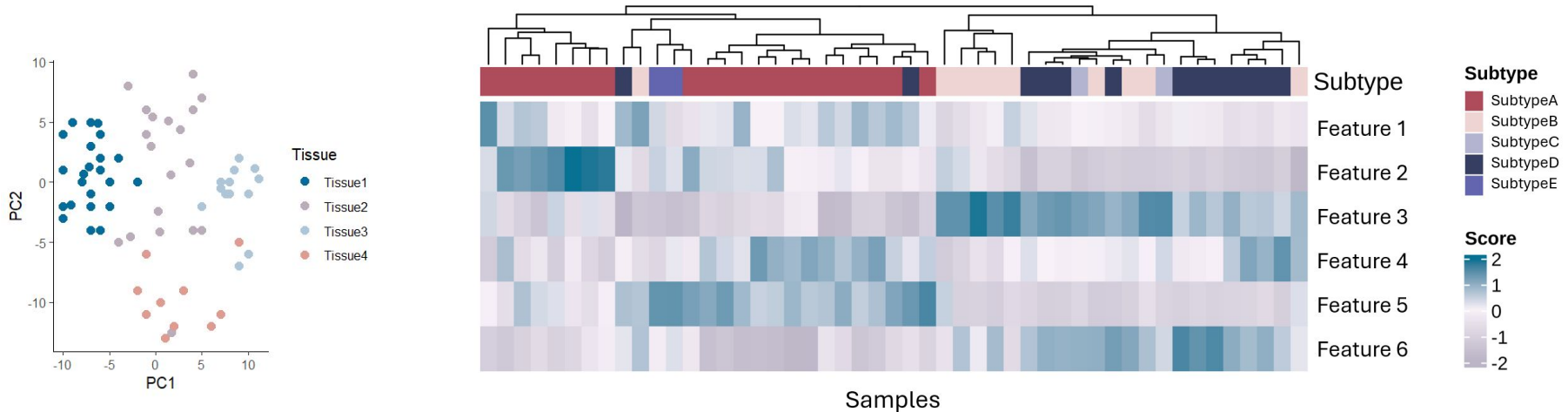
- **Quality Control:** Assess data integrity to check for outliers.
- **Normalization:** Apply techniques such as quantile normalization to ensure comparability across samples.
- **Handling Missing Values:** Use imputation methods or filter out genes/samples with excessive missing data.



Pharmacogenomics Data Analysis Walk-Through

Step 2. Exploratory Data Analysis:

- **Visualization:** Visualize the variance in gene expression data.
- **Clustering:** Implement hierarchical clustering to identify sample groupings.



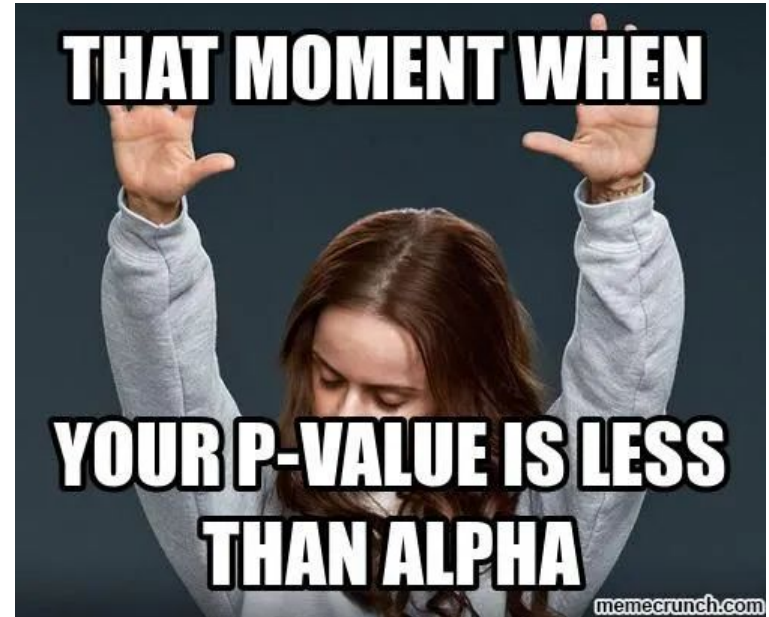
Pharmacogenomics Data Analysis Walk-Through

Step 3. Statistical Testing:

- **Association Tests:** Use association methods to find correlations between gene expression and drug response.

We will explore this in further depth in a few slides!

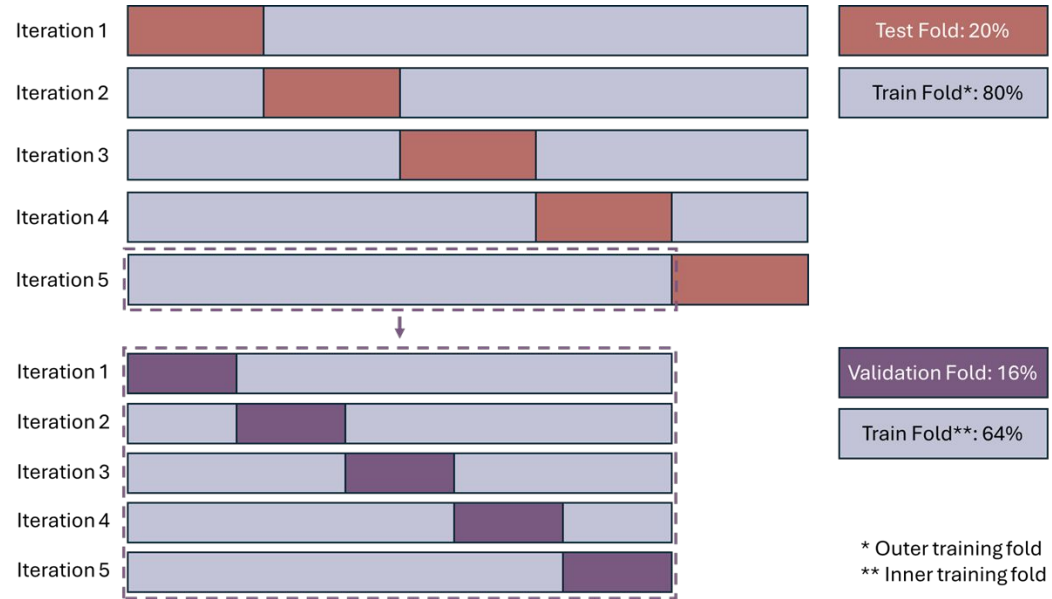
- **Multiple Testing Correction:** Apply methods like Benjamini-Hochberg to control the false discovery rate.



Pharmacogenomics Data Analysis Walk-Through

Step 4. Validation:

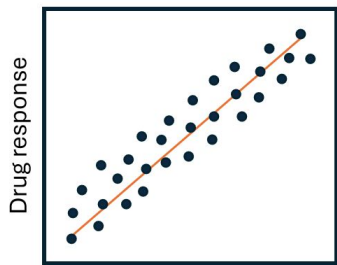
- **Cross-Validation:** Split the data into training and test sets
- **Independent Datasets:** Validate findings in separate datasets.



Understanding Drug Response Associations

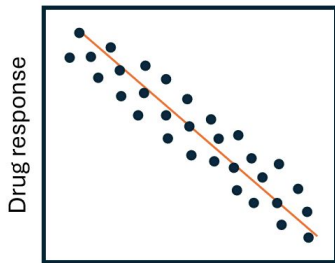
Linear Associations

Positive association



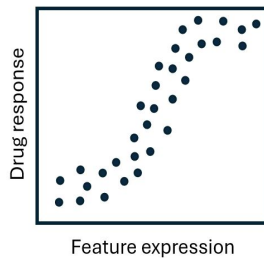
Feature expression

Negative association

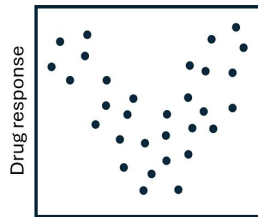


Feature expression

Nonlinear associations

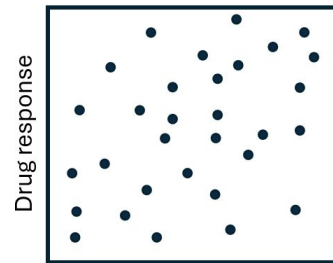


Feature expression



Feature expression

No association

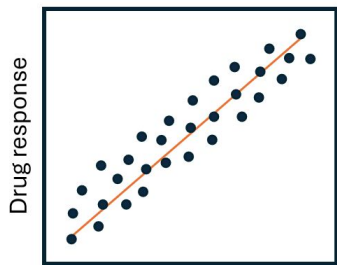


Feature expression

Understanding Drug Response Associations

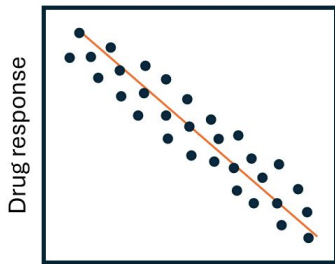
Linear Associations

Positive association



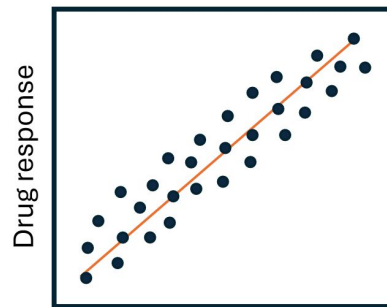
Feature expression

Negative association



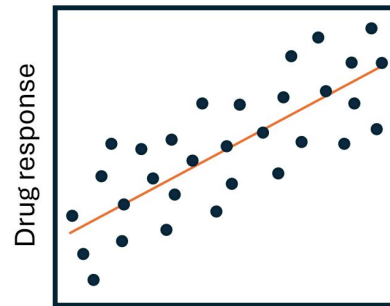
Feature expression

Stronger association



Feature expression

Weaker association



Feature expression

Next: statistical methods for quantifying linear associations

Pearson's Correlation Coefficient

Purpose: measures the strength and direction of a linear relationship between two continuous variables (feature expression and drug response)

Formula:

$$r = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum (X_i - \bar{X})^2 \sum (Y_i - \bar{Y})^2}}$$

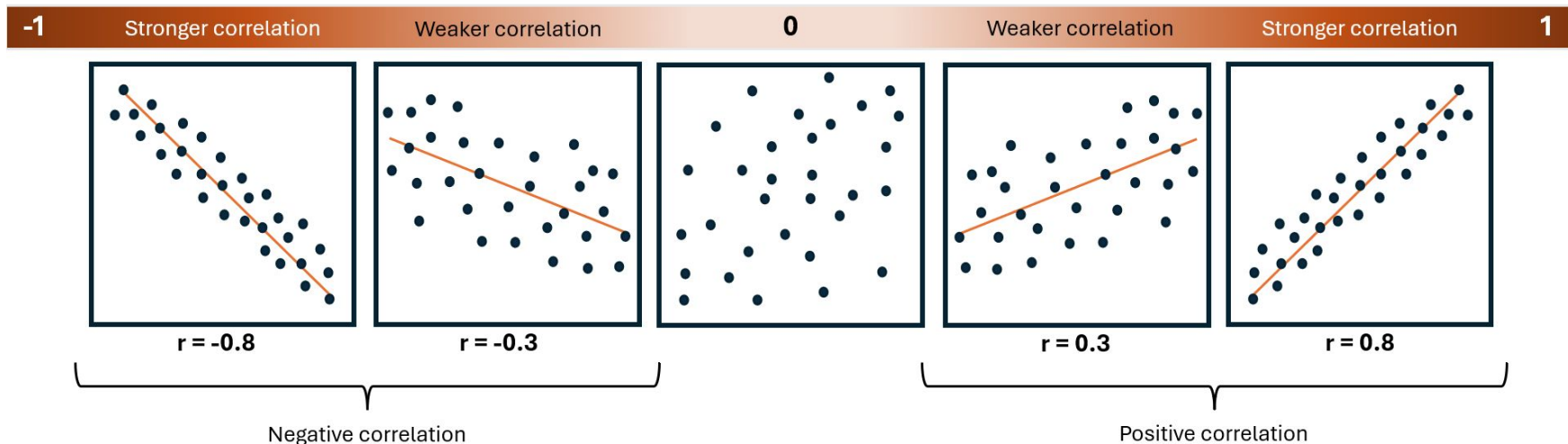
R:

```
c <- cor.test(gene.expr['GeneX', ],  
              drug.response['DrugY', ],  
              method = 'pearson')
```

Python:

- `scipy.stats.pearsonr`
- `pandas.DataFrame.corr(method='pearson')`
- `numpy.corrcoef()`

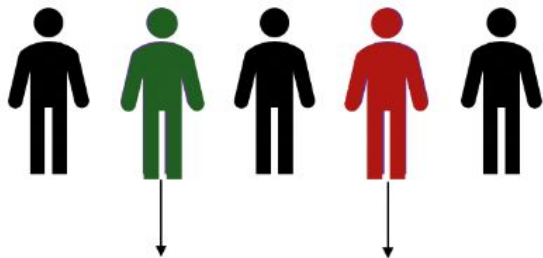
Pearson's Correlation Coefficient



Positive Correlation: Feature expression associated with increased drug sensitivity.
Negative Correlation: Feature expression associated with decreased drug sensitivity.

Concordance Index (C-Index)

Purpose: evaluates the ability of a variable to predict the ranking of a continuous outcome



Feature expression	lower	higher
Response to treatment	Better	Worse

$$\text{C-Index} = \frac{\text{\# concordant pairs}}{\text{total pairs}}$$

```
cindex <- concordance.index(gene.expr['GeneX', ],  
                             surv.time = drug.response['DrugY', ],  
                             surv.event = rep(1, length(drug.response['DrugY', ])))
```


Concordance Index

Purpose: evaluates the ability of a variable to predict the ranking of a continuous outcome

Interpretability:

- **CI = 1** denotes a perfect consistency
- **CI = 0.5** denotes a random association (no consistency)
- **CI = 0** denotes a perfect *inverse* consistency

Consistency: Feature expression associated with increased drug sensitivity.

Inverse consistency: Feature expression associated with decreased drug sensitivity.

Modeling Approaches / Architectures

Univariable

- Considers single feature value as independent variable
- E.g. DNA mutation, gene fusion, mRNA/protein expression, methylated locus



Multivariable

- Multiple independent variables
1. **Linear** combination of features
 - E.g. TMB, MSI, gene expression signatures (OncoType Dx, etc)
 2. **Non-linear**
 - More computationally expensive
 - *May* yield better performance



Multimodal

- Multivariate model with different types of input features / feature spaces
- *May* yield better performance but but could come at the cost of model complexity or interpretability!



Simple predictors

Complex predictors

TMB = tumor mutational burden, MSI = microsatellite instability

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Simple predictors

Complex predictors

TMB = tumor mutational burden, MSI = microsatellite instability

Univariable vs. Multivariable Analysis

Univariable Analysis

```
lm(drug['TAXOL',] ~ expr['TP53', ])  
|> summary()
```

Interpretation:

The coefficient indicates the change in taxol drug response per unit change in *TP53* gene expression.

Multivariable Analysis

```
lm(drug['TAXOL',] ~ expr['TP53', ] +  
  expr['BRCA1', ] + clin[, 'AGE'])  
|> summary()
```

Interpretation:

Coefficients represent the effect of each variable adjusting for others.

Univariable vs. Multivariable Analysis

	Advantages	Limitations
Univariable	• Simple • Easy to interpret	• Ignores potential confounding factors • May miss interactions between variables
Multivariable	• Adjusts for potential confounders • Identifies independent predictors • Can yield more powerful predictions	• Requires larger sample sizes • Requires more compute power (GPUs) • More complex interpretation

Summary and Q&A

Statistical Testing (For Reference)

Parametric:

- Requires normally distributed data
- Involves parameters like mean and standard deviation

Non-Parametric:

- Useful for skewed data or outliers
- Often used for ordinal or nominal data
- Does not rely on parameters; compares medians or ranks

Non-parametric test	Alternative parametric test	Type of input data
Sign test	One-sample Z-test	One group
Wilcoxon signed rank test	One sample Z-test	One group
Rank-sum (Mann-Whitney U-test)	Unpaired t-test	Two groups –unpaired
Wilcoxon signed rank test	Paired t-test	Two groups –paired
Kruskal-Wallis	One-way ANOVA	Multiple groups
Spearman correlation	Pearson correlation	Two groups –paired

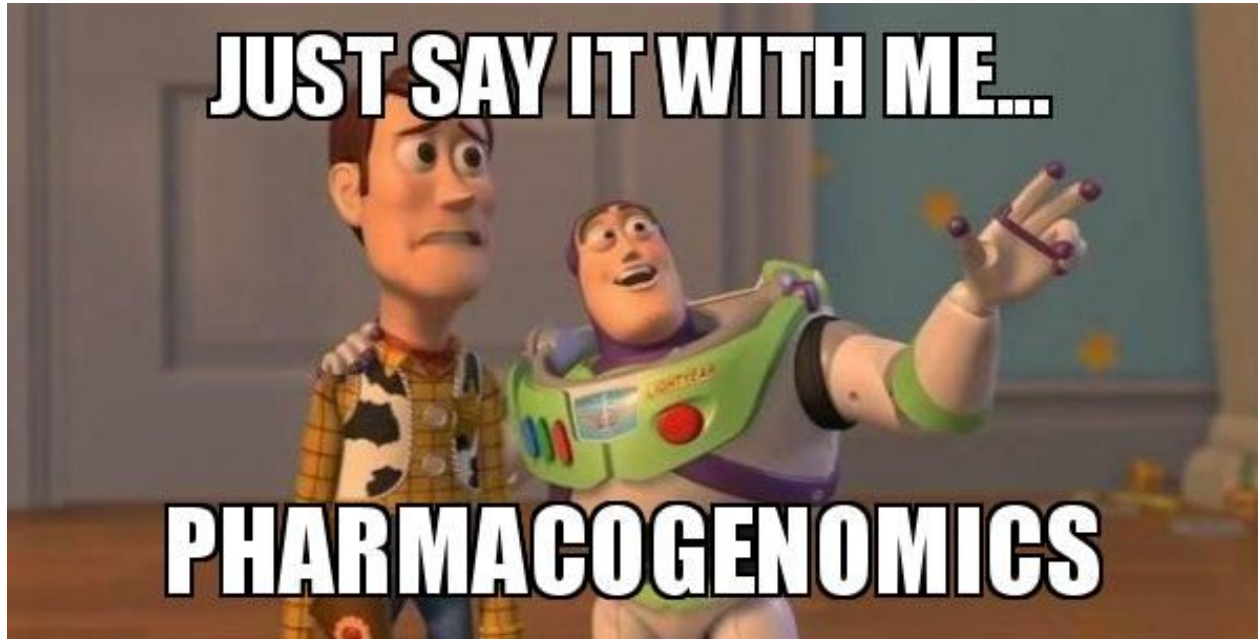
Pharmacogenomics (For Reference)

Topic	Key Points	Examples
Modeling Approaches	Univariable, Multivariable, and Multimodal models. Increasing complexity improves performance but adds computational cost.	DNA mutations, TMB, MSI, gene expression signatures, multimodal AI models.
Univariable Models	Uses a single feature as a predictor. Simple but may miss interactions.	Single-gene mutations (e.g., EGFR, BRAF), protein expression.
Multivariable Models	Uses multiple independent variables. Can be linear or non-linear.	TMB + MSI for immunotherapy response, OncoType Dx.
Multimodal Models	Integrates different data types (genomics, imaging, clinical). High complexity, may improve prediction.	AI models combining sequencing, imaging, and patient history.

Pharmacogenomics (For Reference)

Topic	Key Points	Examples
Feature Selection	Choosing the most informative variables to improve model performance.	LASSO, PCA, biological relevance-based filtering.
Predictive Biomarkers	Biomarkers that help predict drug response. Essential in precision medicine.	HER2 for breast cancer therapy, PD-L1 expression for immunotherapy.
Response Prediction	The goal of pharmacogenomics models—predicting who benefits from treatment.	Tumor sequencing to guide chemotherapy selection.

Questions?



Supplemental

Modeling Approaches / Architectures

Univariable

- Considers single feature value as independent variable
- E.g. DNA mutation, gene fusion, mRNA/protein expression, methylated locus



Multivariable

- Multiple independent variables
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- *May* yield better performance but but could come at the cost of model complexity or interpretability!



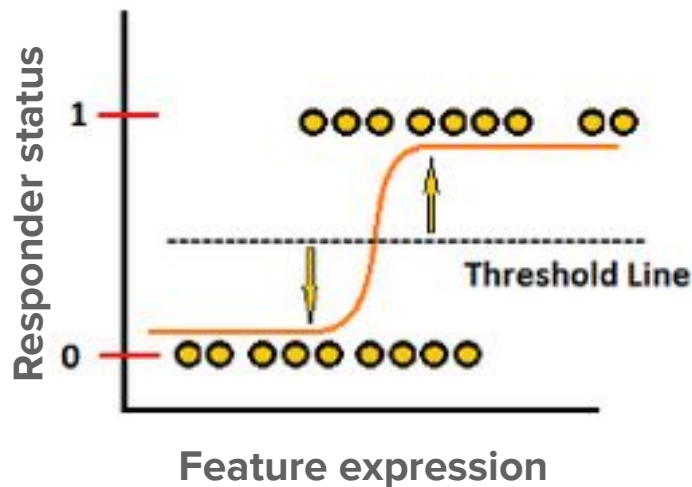
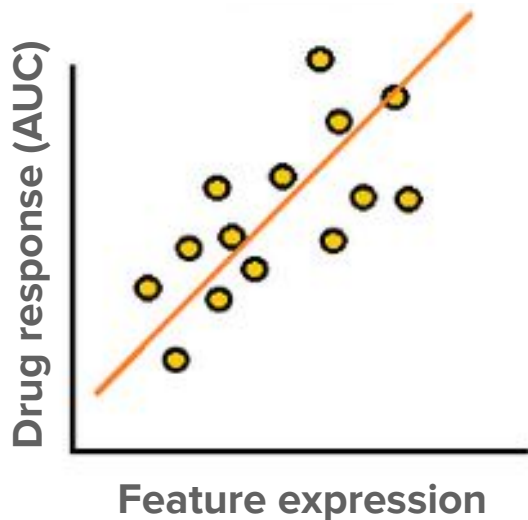
Simple predictors

Complex predictors

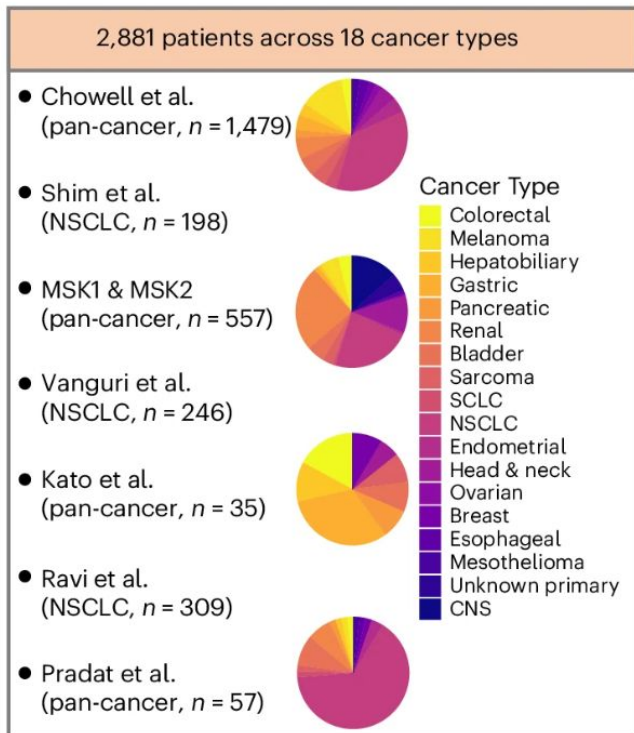
TMB = tumor mutational burden, MSI = microsatellite instability

ML for Drug Response Predictions

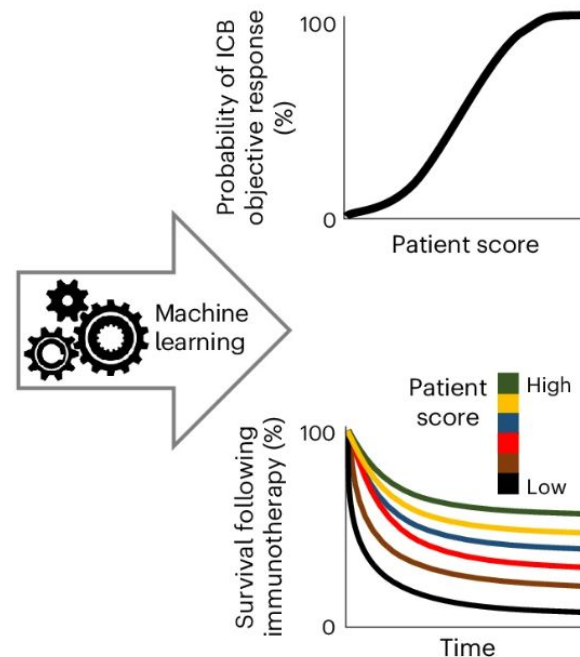
- **Linear regression** → e.g., predicting cell line drug response
- **Logistic regression** → e.g., predicting patient response to a drug



LORIS: Predicting immunotherapy response with logistic regression

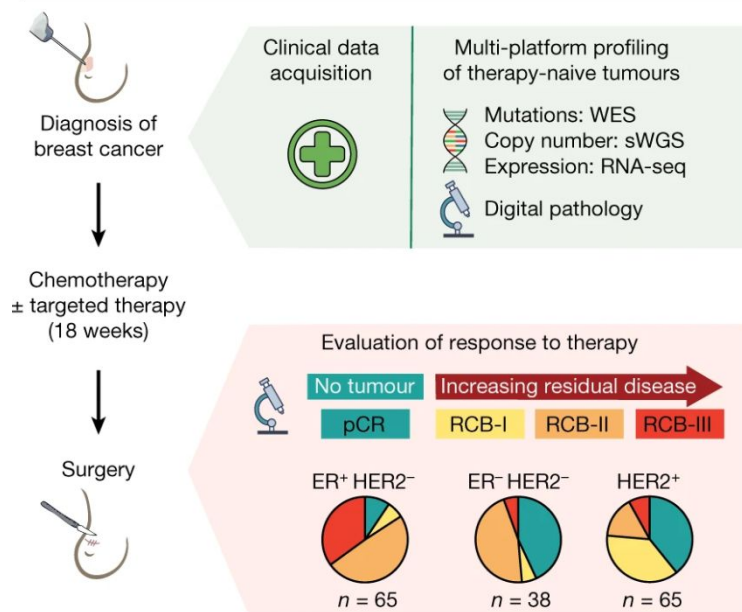
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Features
- TMB - PD-L1 TPS - Systemic therapy history - Blood albumin level - Blood NLR - Patient age - Patient sex - ICB drug class - Cancer type ...
Patient outcomes
- ICB objective response (training target) - Overall survival - Progression-free survival

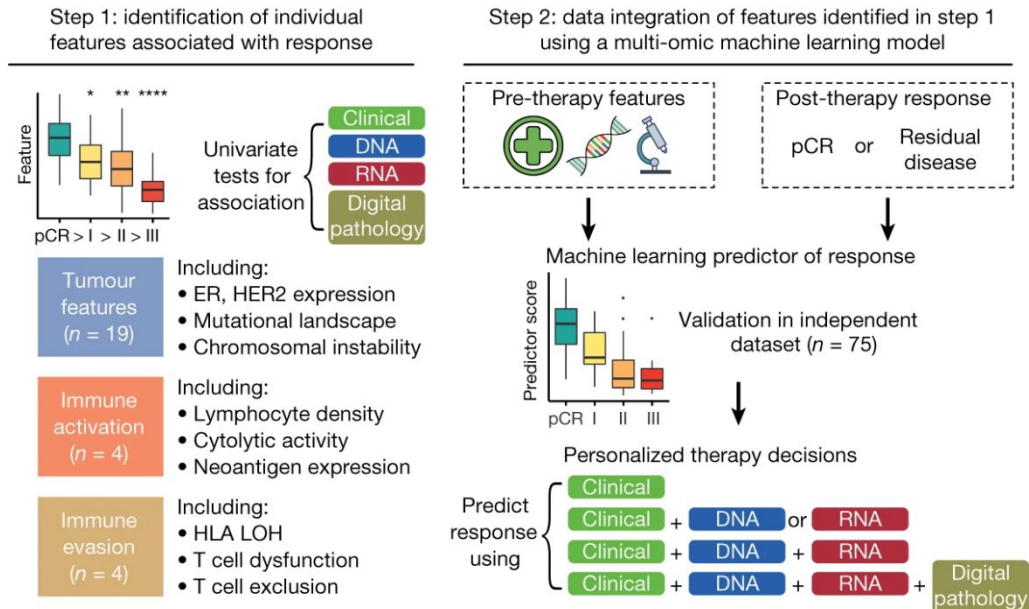


Multi-omic ML predictor of therapy response

Clinical work flow and data acquisition

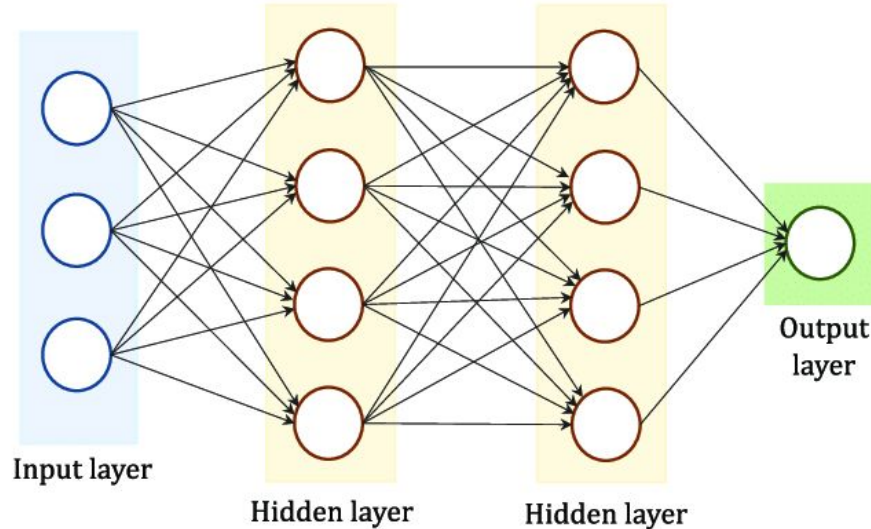


Prediction of response to neoadjuvant therapies



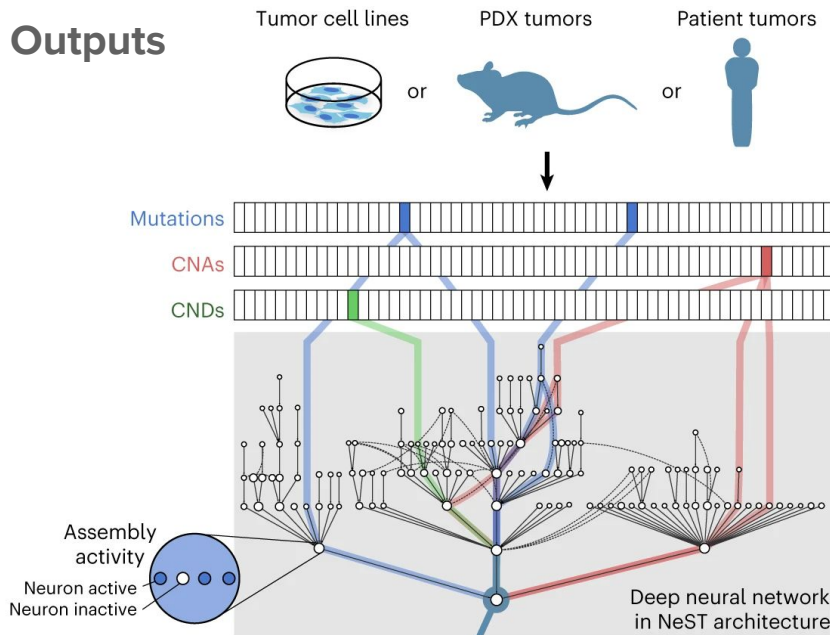
Deep Learning for Drug Response Predictions

- **Deep Learning:** machine learning based on neural networks
- **Neural networks:** models inspired by the biological neural network of our brains



NestVNN: A Visible Deep Learning Model

Model Outputs



Predicted drug response

